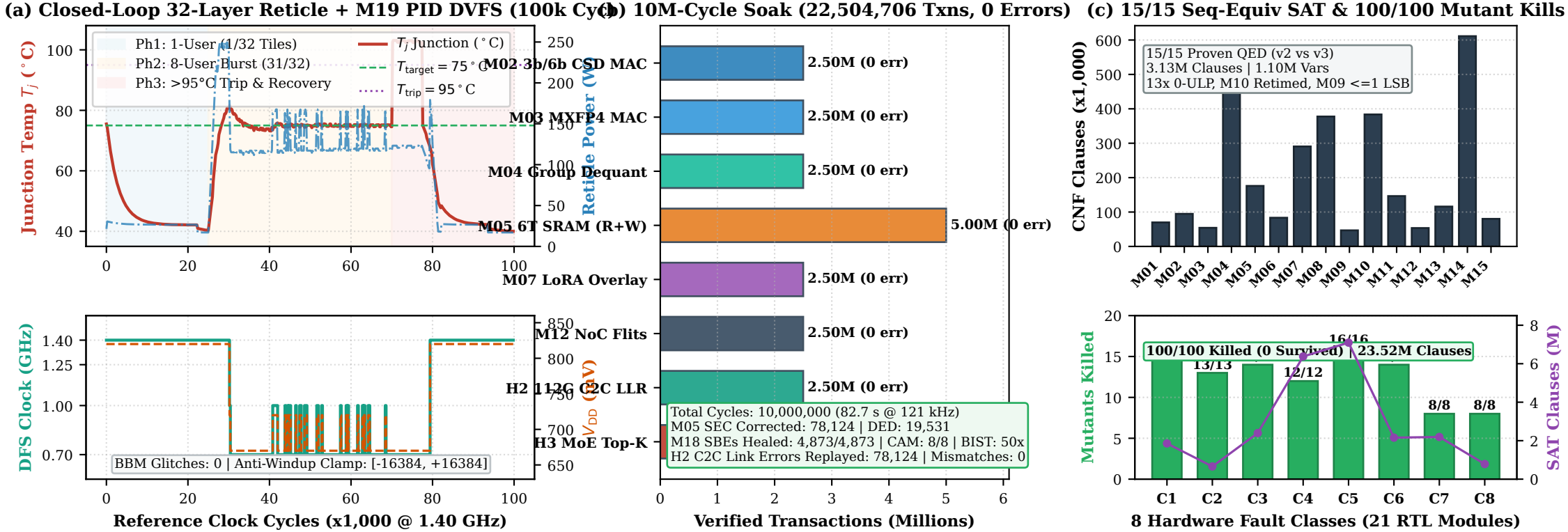
(a) Closed-Loop 32-Layer Reticle + M19 PID DVFS (100k Cycles)
Ph1: 1-User (1/32 Tiles)
Ph2: 8-User Burst (31/32)
Ph3: >95°C Trip & Recovery
T_j Junction (°C)
T_{target} = 75 °C
T_{trip} = 95 °C
Junction Temp T_j (°C)
Retice Power (W)
DFS Clock (GHz)
V_{DD} (mV)
BBM Glitches: 0 | Anti-Windup Clamp: [-16384, +16384]
Reference Clock Cycles (x1,000 @ 1.40 GHz)
(b) 10M-Cycle Soak (22,504,706 Txns, 0 Errors)
M02 3b/6b CSD MAC
M03 MXFP4 MAC
M04 Group Dequant
M05 6T SRAM (R+W)
M07 LoRA Overlay
M12 NoC Flits
H2 112G C2C LLR
H3 MoE Top-K
2.50M (0 err)
5.00M (0 err)
Total Cycles: 10,000,000 (82.7 s @ 121 kHz)
M05 SEC Corrected: 78,124 | DED: 19,531
M18 SBEs Healed: 4,873/4,873 | CAM: 8/8 | BIST: 50x
H2 C2C Link Errors Replayed: 78,124 | Mismatches: 0
Verified Transactions (Millions)
(c) 15/15 Seq-Equiv SAT & 100/100 Mutant Kills
15/15 Proven QED (v2 vs v3)
3.13M Clauses | 1.10M Vars
13x 0-ULP, M10 Retimed, M09 <=1 LSB
CNF Clauses (x1,000)
M01 M02 M03 M04 M05 M06 M07 M08 M09 M10 M11 M12 M13 M14 M15
100/100 Killed (0 Survived) | 23.52M Clauses
13/13
12/12
16/16
8/8
8/8
Mutants Killed
SAT Clauses (M)
C1 C2 C3 C4 C5 C6 C7 C8
8 Hardware Fault Classes (21 RTL Modules)